

Markowitz-based portfolio selection with minimum transaction lots, cardinality constraints and regarding sector capitalization using genetic algorithm

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ABSTRACT

Heuristic algorithms strengthen researchers to solve more complex and combinatorial problems in a reasonable time. Markowitz's Mean-Variance portfolio selection model is one of those aforesaid problems. Actually, Markowitz's model is a nonlinear (quadratic) programming problem which has been solved by a variety of heuristic and non-heuristic techniques. In this paper a portfolio selection model which is based on Markowitz's portfolio selection problem including three of the most important limitations is considered. The results can lead Markowitz's model to a more practical one. Minimum transaction lots, cardinality constraints (both of which have been presented before in other researches) and market (sector) capitalization (which is proposed in this research for the first time as a constraint for Markowitz model), are considered in extended model. No study has ever proposed and solved this expanded model. To solve this mixed-integer nonlinear programming (NP-Hard), a corresponding genetic algorithm (GA) is utilized. Computational study is performed in two main parts; first, verifying and validating proposed GA and second, studying the applicability of presented model using large scale problems.

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1. Introduction

Markowitz's modern portfolio theory has made a new paradigm of portfolio selecting for investors in order to form a portfolio with the highest expected return at a given level of risk tolerance (the lowest level of risk tolerance at a given expected return) (Markowitz, 1952; Markowitz, 1959; Oh, Kim, Min, & Lee, 2006). Lots of efforts have been performed by experts in order to solve and expand Markowitz's model. These attempts, regarding the limitations of a factual market, have tried to make his model more practical.

In 1956, Markowitz represented the critical line method to solve his quadratic model (Markowitz, 1956). Wolfe tried to solve Markowitz's model by Simplex algorithm (Wolfe, 1959). Konno's new definition of risk in his mean absolute deviation (MAD) model, has been led several investigations. Interestingly, Konno's model can be solved by linear methods like Simplex, (Konno, 1990; Konno & Yamazaki, 1991). Again, Markowitz, himself, studied more complex objective functions, based on the notions of semi-variance (Markowitz, Todd, Xu, & Yamane, 1993).

In 1993, Speranza introduced a more general model with a weighted risk function (Speranza, 1993) and also proposed a

mixed-integer programming, considering realistic characteristics in portfolio selection, such as minimum transaction lots and maximum number of securities. His researches have been always based on MAD model of Konno not on Markowitz's (Speranza, 1996). Other investigations added some other constraints; Yoshimoto considered multi period portfolio selection with transaction costs based on Markowitz's (Yoshimoto, 1996). Speranza and Mansini's research, regarded transaction costs with and without minimum transaction lots but again not based on Markowitz's model (Mansini, & Speranza, 1997). In 2001, Konno proposed an algorithm for his portfolio optimization problems regarding transaction costs and minimum transaction lots (Konno, & Wiyayanayake, 2001).

In the field of model solving, Arnone presented a Genetic Algorithm for an unconstrained portfolio optimization problem (Arnone, Loraschi, & Tettamanzi, 1993), but the first use of genetic algorithm for Markowitz's model (without any extra constraints) was done by Shoaf (Shoaf, & Foster, 1996). Rolland utilized Tabu search (TS) to solve Markowitz's (Rolland, 1997). Later, to corroborate the necessity and desirability of heuristic algorithms, Mansini and Speranza proved that the portfolio selection problem with minimum transaction lots is an NP-complete problem. Subsequently, they proposed three heuristic algorithms to figure out the MAD model of Konno (Mansini, & Speranza, 1999). Afterwards, they (with Kellerer) extended their model in order to take fixed transaction costs into account (Kellerer, Mansini, & Speranza, 1999).

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At the first years of 20th century, some studies threw more light on the capabilities of heuristic algorithms in portfolio selection problems; Xia, Orito, Streichert, Fieldsend and Lai in showing the excellent performance of GA, Gilli in utilizing Simulated Annealing (SA), and Chang's group experimented a variety of meta heuristics, including SA, TS and GA for portfolio selection model without trading and turnover constraints. Although Chang mentioned that GA is best able to approximate the Unconstrained Efficient Frontier (UEF) but he could not found any individual heuristic performed better than the other ones (Chang, Meade, Beasley, & Sharaiha, 2000; Fieldsend, Matatko, & Peng, 2004; Gilli, & Kellezi, 2000; Lai, Yu, Wang, & Zhou, 2006; Orito, Yamamoto, & Yamazaki, 2003; Streichert, Ulmer, & Zell, 2003; Xia, Liu, Wang, & Lai, 2000). Schaerf first improved Chang's article, thereafter, in another research he proposed a Tabu Search algorithm in order to solve Markowitz's model with cardinality of the portfolio (Schaerf, 2002a,b). Lin considered the multi objective genetic algorithm for portfolio selection problem (Lin, Wang, & Yan, 2001). Crama utilized a Simulated Annealing to solve Markowitz's model with cardinality and Turn Over constraints (Lin et al., 2001).

Oh proposed a new portfolio selection algorithm based on portfolio beta and regarding sector capitalization, he profited genetic algorithm as his solver (Oh, Kim, & Min, 2005). Stein exploited Meta heuristics for Markowitz's model with cardinality and Buy-in thresholds (Stein, Branke, & Schmeck, 2005). Finally, in 2007, Markowitz's model and heuristic algorithms are still taken into considerations; Lin and Liu (2007) considered Markowitz's model with minimum transaction lots and they presented three other models. They also utilized GA to solve their proposed models. Fernández and Gómez (2007) paid attention to Markowitz's model with cardinality and bounding constraints. They utilized neural networks as their solver.

On the other side, genetic algorithms, roughly speaking, are one of the most popular heuristic optimization techniques were originally developed by Holland (1975, 1992). Subsequently, the GA was applied to optimization problems in biology, engineering, and operations research. In ninety decade, different aspects of GA were expanded; Hartl researched on convergence of GA (Hartl, 1990), Radcliffe and Bean studied on cross over operator (Bean, 1992; Radcliffe, 1991, 1992), Miller and Goldberg (1995) researched on selection strategies, and finally, Vose and Koza (Koza by developing Genetic Programming), investigated on whole concept of basic GA of Holland (Koza, 1994; Vose, 1991). More information about GAs and also Evolutionary Algorithms can be found in Reeves, Mitchell and Baeck's researches (Baeck, Fogel, & Michalewicz, 1997; Mitchell, 1996; Reeves, 1993).

In this paper, a portfolio selection model which is based on Markowitz's portfolio selection problem including three of the most important limitations is considered. The results can lead Markowitz's model to a more practical one. Minimum transaction lots, cardinality constraints (both of which have been presented before) and market (sector) capitalization (which is proposed in this research for the first time as a constraint for Markowitz model), are taken into account in presented model. In particular, in Section 2 the expanded model is represented and described. In Section 3 the characteristics of proposed GA is explained. Empirical experiments which research the performance of proposed model and also the GA are discussed in Section 4. Finally, in Section 5 conclusions are remarked and possible future researches are provided.

2. The proposed model

In this section the model will be described. It has been mentioned that the proposed model is based on Markowitz's mean-variance portfolio selection model which doesn't consider the situ-

ation of a real market as cardinality constraint and minimum transaction lots. In order to explain the model let:

N	be the total number of securities available
r_i	be the expected return of security "i" ($i = 1, \dots, N$)
σ_i	be the covariance between securities "i" and "j" ($i = 1, \dots, N; j = 1, \dots, N$)
R	be the minimum desired expected return (of investor)
B	be the total available budget
w_i	be the weight of security "i" in portfolio
c_i	be the price of security "i" multiply by minimum transaction lots (price of Lot i)
x_i	be the number of minimum transaction lots of Lot "i" which model suggest to purchase
B_{upper_i}	be the maximum permitted amount of investing on security i
B_{lower_i}	be the minimum permitted amount of investing on security i
z_i	be a binary variable which will be 1 if any of security "i" ($i = 1, \dots, N$) is held and 0 otherwise
M	be a large number
K	be the desired number of securities in the portfolio
S	be the total number of available sectors
y_j	be a binary variable which will be 1 if any security of sector "j" is held and 0 otherwise
i	be the index of securities
j	be the index of sectors

Then the extended model will be

$$Z = \min \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij}, \quad (1)$$

$$\sum_{i=1}^N x_i c_i r_i \geq bR, \quad (2)$$

$$\sum_{i=1}^N x_i c_i \leq b, \quad (3)$$

$$w_i = \frac{c_i x_i}{\sum_{i=1}^N c_i x_i}, \quad i = (1, 2, \dots, N), \quad (4)$$

$$\sum_{i=1}^N z_i = k, \quad (5)$$

$$B_{lower_i} - M(1 - z_i) \leq x_i c_i \leq B_{upper_i} + M(1 - z_i), \quad i = (1, 2, \dots, N), \quad (6)$$

$$\frac{x_i}{M} \leq z_i \leq x_i, \quad i = (1, 2, \dots, N), \quad (7)$$

$$\frac{\sum_{i \in j} z_i}{M} \leq y_j \leq M \sum_{i \in j} z_i, \quad j = (1, 2, \dots, s), \quad (8)$$

$$\sum_{i \in [j]} w_i + (1 - y_j) \geq \sum_{i \in [j+1]} w_i, \quad j = (1, 2, \dots, s-1), \quad (9)$$

z_i and y_j are binary variables; x_i is an integer variable and over zero (no short sale is allowed).

As it can be observed above, the extended model is formulated. Eq. (1) is the objective function which determines the risk. Definitely, it should be minimized. Eq. (2) ensures achieving the desired minimum expected return of investor. Eq. (3) considered budget limitation. Finally, the weights are calculated in Eq. (4).

Till the forth equation, the practical constraint of minimum transaction lots is considered by defining variable c_i as the price of security "i" multiply by minimum transaction lots. It means c_i is the price of purchasing one minimum transaction lot of a security i. This method is mentioned in Mansini and Speranza's study (Mansini & Speranza, 1997). Consequently, x_i represents the number of minimum lots, so, $x_i c_i$ indicates the part of the b (total

available budget) that the model suggests to investor to invest in security i .

In order to extend the model for covering cardinality constraints, the fifth, sixth and seventh constraints (Eqs. (5)–(7)) should be considered. The fifth equation ensures that exactly “ k ” securities will be held in suggested portfolio. Purchasing limitations which doesn’t let “ $x_i c_i$ ” be extremely high or low are considered by Eq. (6). Indeed it limits the proportion of budget which is invested in security i . Definitely, B_{lower} and B_{upper} depend on the total available budget. Finally, regarding Eq. (7), binary variable z_i will be 1 if any of security “ i ” ($i = 1, \dots, N$) is held and 0 otherwise.

2.1. Regarding sector capitalization

The significance of market capitalization is discussed in Chang’s article and also in Oh’s (Chang et al., 2000; Oh et al., 2006). Investors tend to invest in sectors with higher value to reduce their risk of investment. In this proposed model, sector capitalization is considered for the first time as a constraint to make the selected security more reasonable and safer.

Regarding Eq. (8), if no security is chosen from sector “ j ”, the related z_j will be zero and y_j will be zero too. But if even one security in sector “ j ” is selected, y_j should be more than zero (actually it should be more than $\frac{\sum_{i \in j} z_i}{M}$) and because it is a binary variable, it will be “1”.

Regarding market capitalization, it is determined if some securities in different sectors are selected, the total weight of each selected stock should be related with its sector capitalization. Roughly speaking, the sector with more capitalization should have more proportion in ultimate portfolio if it is selected. Therefore, it is an important point that this limitation is disabled when no security is chosen from a sector.

For example if sector 1 has more capitalization than sector 2 and there are some securities in portfolio from both sectors, the total weight of securities in sector 1 should be more than the total weight of securities in sector 2. But if there is no securities in portfolio from sector 1, this limitation will be disabled.

Eq. (9) covers above explanations. As it is presented in Eq. (9), it is assumed (as putting index “ j ” in brackets) that sectors are sorted regarding market capitalization. It means after sorting, sector 1 has more capitalization than sector 2 and so on.

If no security of sector “ j ” is selected, y_j will be zero and because of “1” which remains in the left side of Eq. (9), and regarding this point; $\sum_{i \in j} w_i \leq 1$, the limitation disregarding to the weights will never violates otherwise the limitation have to be applied.

To solve the proposed model, genetic algorithm is applied. The procedure of the proposed GA will be explained in Section 3.

3. Utilizing genetic algorithm

Genetic Algorithms which were originally developed by Holland in 1975, is one of the most popular heuristic optimization techniques (Holland, 1975, 1992). GAs originate from the famous Darwin’s principal of natural selection and the well-known phrase; “survival of the fittest”. Actually, it can explore the solution space by using the concept taken from natural genetics and evolutionary theory. In Genetic Algorithms, an initial population which consists of several feasible solutions (chromosomes) is selected randomly as the first generation, and then parents are chosen from this population and produce child (offspring). Consequently, the second generation is a combination of the parents and their offspring. The cross over and mutation operators interfere in the process of “generation producing” and in any generation the desirability of solutions (chromosomes) are calculated by fitness function which is originated from the objective function of the model.

The basic steps of a simple GA could be (Chang et al., 2000):

Generate an initial population

Evaluate fitness of individuals in the population

Repeat:

Select parents from the population (**regarding all constraints**)

Recombine (mate) parents to produce children using cross over and mutation operators

Evaluate fitness of the children

Replace some or all of the population by the children

Until a satisfactory solution has been found.

About the proposed GA of this investigation, two points should be mentioned; first the cross over operator is almost based on the RAR operator of Radcliffe (Radcliffe, 1991), and second the selection strategy is as follow:

- The fitness of individuals in the population are evaluated.
- Then the population is sorted regarding fitness value.
- Half of the fittest will be in the next generation.
- Parents are selected randomly.
- Each parent creates one child using cross over and mutation operators.
- These offspring, who are in the number of half of the population, proceed to the next generation. So, finally, half of the fittest plus the generated offspring, will form the next generation.

About the cross over operator, some points should be mentioned. Regarding aforesaid indications, cross over operator is in some parts based on RAR operator of Radcliffe. The rules are also so simple; if both of parents contain a special stock (the number is not important), then that stock will be assigned to the portfolio of produced child and the number of that stock will randomly be assigned between the values of two parents. Vice versa; if both of parents are in the same situation of not having a special stock, then that stock will definitely not be in the portfolio of generated child. If one of the parents has a special stock which the other one has not, then final value will randomly be assigned between these two. Meanwhile, dynamically, all of the constraints monitor the characteristics of generated child and if it is not feasible, the process of cross over will be repeated.

In Section 4 this GA which is coded by MATLAB 7 will be verified and the aspects of the GA and the model will be more clarified.

4. Empirical experiments

The empirical experiments are organized in four parts; first the parameters of the proposed GA are set, second to verify the logical reactions of proposed GA, data are retouched intentionally and the results are analyzed, third in order to validate proposed GA (which is coded by MATLAB 7), it is compared with model results in LINGO which can generate global optima for small problems (but because of LINGO’s limitation, this comparison couldn’t be performed in large scale data) and finally, the performance of the proposed GA of extended model is verified by using two generated markets (randomly by Matlab) with 500 and 2000 stocks.

4.1. Parameters setting

Some experiments are performed to clarify the best parameters of the proposed GA. The results show us our GA is converged enough in more than 500 iterations and the best probabilities of cross over and mutation rates are 1, 0.5, respectively. Population sizes perform acceptable in 20 and 30.

It is not necessary to review all of the experiments in this section except about population size and it is because of its importance. This experiment is performed with a 150-stock-problem with different values of “population size” (6, 10, 16, 20, 30, 50 and 100). Two criteria are recorded; objective function value and time of solving. Other parameters of the GA including iteration, R , cross over and mutation probabilities are; 1000, 5, 1, 0.5 respectively and $R = 5$. The results are represented in Table 1, (Table 2) and (Fig. 1). In (Tables 1 and 2), best objectives are average of four times run. In (Fig. 1), best objectives are multiplied by 100 in order to make the chart clearer.

By analyzing (Tables 1 and 2) and (Fig. 1), the stability of best objective is distinguishable with “population size” of 20, 30 and more. But regarding the shift of time by using more than 30 “population size”, it can be concluded that 20 and 30 are the most suitable and efficient choices for parameter of “population size”.

Other parameters are assigned during other types of experiments which are not necessary to be pointed.

4.2. Sensitivity analysis

As it was expected, some special studies showed us the precision of the proposed GA. For example in a generated problem with four stocks and two sectors (sector “1” and “2”, two stocks in each sector), the stocks in the sector 1 has more market capitalization than the other sector but they are assigned very undesirable (with high variances and low returns). Then a portfolio with 3 stocks are asked (so GA has to select from those undesirable ones), the results show us the total weight of sector “2” become 0.49999 and the total weight of sector “1” become 0.50001, because regarding sector capitalization, GA is forced to assign the total weights of sector “1” more than the total weights of sector “2”. It means the proposed GA of the model performs the most logical reaction which is expected. Some other experiments are performed to make us sure about the results of GA. For example some of are our experiments were:

- The different solutions in different runs (definitely with same determined parameters).
- How the GA uses of total available budget.
- Convergence of proposed GA and etc.

These studies show us the suitable performance of the proposed GA so our main empirical studies could be done thoroughly confident.

4.3. Comparing GA and LINGO

In order to verify our proposed GA, a problem of nine stocks in three sectors with sorted sector capitalization is selected. The proposed GA is coded by MATLAB 7 software (Data are linked from MS Excel 2007) and the experiments are accomplished by a PC with an Intel Pentium IV (Celeron, 2.8 GHZ) processor under Windows XP using 512 MB of RAM. Because of LINGO's limitations, just a 9-stock-problem is selected; for instance, in a 30-stock-problem which is solved by the proposed GA in a short time (around 4 min), with the iteration of 1000 and the best objective of 0.0132 ($k = 5$, population size = 30, cross over and mutation probability = 1, .5, respectively), LINGO cannot reach to any local optimum better than 0.0213 in a very long time (around 24 h) and with the iteration over 2,000,000 (and actually it did not stop). Therefore the mentioned problem is selected which is solved in a short reasonable time by both of LINGO and GA and the proposed GA could be validated. This method can be found in Crama, Xia and Lin's articles (Crama, & Schyns, 2003; Lin and Liu, 2007; Xia et al., 2000).

The results of the experiments are presented in Table 1–3. In all runs population size is 30, cross over and mutation probabilities are 1, .5, respectively and the number of iterations in MATLAB 7 are 1000.

The experiments for $k = 2, 3$ and 5 and for expected return of 5–12 are examined, while the average return and variance of all stocks are 11% and 8.3%, respectively. The results show us differences between LINGO's global optimum and GA's best objective

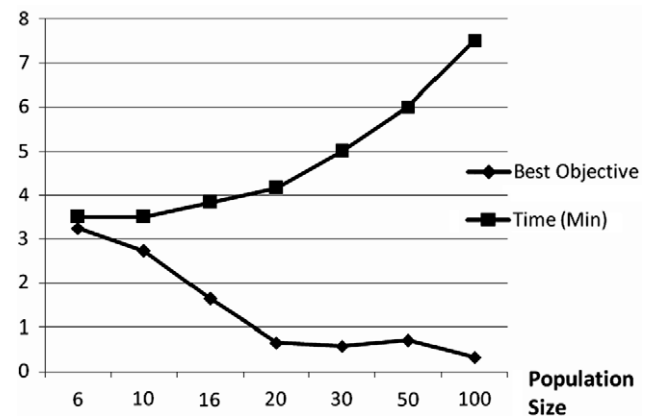


Fig. 1. Population size and its effects on best objective and time of solving.

Table 1
Population size and its effects on best objective and time of solving

Population size	6		10		16		20	
	Best objective	Time (min)	Best objective	Time (min)	Best objective	Time (min)	Best objective	Time (min)
$k = 3$	0.0664	3.5	0.0734	3.5	0.02	3.5	0.0149	4
$k = 4$	0.0085	3.5	0.0011	3.5	0.0172	4	0.0032	4
$k = 5$	0.0223	3.5	0.0074	3.5	0.012	4	0.0016	4.5
Average	0.0324	3.5	0.0273	3.5	0.0164	3.83333	0.00657	4.16667

Table 2
Population size and its effects on best objective and time of solving

Population size	30		50		100	
	Best objective	Time (Min)	Best objective	Time (Min)	Best objective	Time (Min)
$k = 3$	0.0137	4.5	0.0078	5	0.0037	6.5
$k = 4$	0.0022	4.5	0.0122	5	0.0025	7
$k = 5$	0.0016	6	0.0013	8	0.0036	9
Average	0.00583	5	0.0071	6	0.00327	7.5

Table 3The results of experiments in LINGO and the proposed GA for $k = 2$

R	LINGO	GA	Difference
5	0.0000001688	0.0000074850	0.0000073162
6	0.0000001961	0.0046000000	0.0045998039
7	0.0000000031	0.0003586800	0.0003586769
8	0.0000000650	0.0117000000	0.0116999350
9	0.0000008759	0.0000318530	0.0000309771
10	0.0000000002	0.0117000000	0.0116999998
11	0.0000000089	0.0055000000	0.0054999911
12	0.0000003849	0.0021000000	0.0020996151
Average			0.0044995394

are just 0.45%, 0.64% and 7.8% with the average of 2.9%. Such a small difference which is below than 3% can be acceptable and reasonable. So, the results encourage us to use the proposed GA of the model in a factual (large) market. The applicability of proposed GA in real market with large number of stocks (N) is verified in Section 4.4.

4.4. Applicability in large scale problems

To verify capabilities of proposed GA, two problems with 500 and 2000 stocks (N) with 10 sectors are selected, then solving time and best objectives are recorded. The results are presented in Table 4 (The parameters of GA are; population size = 30, cross over and mutation probabilities are 1, .5, respectively). The results show the proposed GA of the model is really reliable. While average return and variance of market are around 7% and 15%, respectively, the appreciable performance of GA in achieving to an acceptable objective in a reasonable time (less than 7 min, despite performing by a Celeron processor) is cleared. Our experiments can show that this model and the proposed GA are applicable in real market with large number of stocks.

Finally, our results can lead us to this conclusion which even in the real markets with real number of stocks, the model with its proposed GA is applicable and it can reach to reasonable objective

Table 4The results of experiments in LINGO and the proposed GA for $k = 3$

R	LINGO	GA	Difference
5	0.0068832220	0.0070000000	0.0001167780
6	0.0068832210	0.0155000000	0.0086167790
7	0.0068833860	0.0085000000	0.0016166140
8	0.0068833860	0.0094000000	0.0025166140
9	0.0068833860	0.0132000000	0.0063166140
10	0.0050166170	0.0082000000	0.0031833830
11	0.0050166170	0.0069000000	0.0018833830
12	0.0068832220	0.0341000000	0.0272167780
Average			0.0064333679

Table 5The results of experiments in LINGO and the proposed GA for $k = 5$

R	LINGO	GA	Difference
5	0.0369502500	0.0971000000	0.0601497500
6	0.0369502500	0.2780000000	0.2410497500
7	0.0369502500	0.1340000000	0.0970497500
8	0.0369502500	0.1729000000	0.1359497500
9	0.0369502500	0.1085000000	0.0715497500
10	0.0369502500	0.0812000000	0.0442497500
11	0.0369502500	0.0812000000	0.04424975
12	0.0870655800	0.2590800000	0.17201442
Average			0.0777453338

Table 6

The applicability of the proposed GA of the model in real markets

Number of stocks	Iteration	k	R	Time (min)	Objective
500	1000	3	5	6	0.0157
500	1000	3	7	7	0.0025
500	1000	3	9	7	0.017
500	1000	5	5	6	0.0216
500	1000	5	7	7	0.0069
500	1000	5	9	8	0.0201
2000	500	3	5	16	0.1136
2000	500	3	7	16	0.0877
2000	500	3	9	16	0.0241
2000	500	5	5	16	0.0628
2000	500	5	7	17	0.0405
2000	500	5	9	17	0.0161
Average	–	–	–	11.583	0.035717

function (average of 0.035716667 means having just 3.5% risk) in a reasonable time (7 min). Roughly speaking, these results show us; not only the proposed GA of our model can achieve to reasonable objective function near to Global Optima, but also, it is applicable from both aspects of time and reliability of best objectives in real markets (see Table 5).

5. Conclusions and future researches

In this paper a portfolio selection model based on Markowitz's mean–variance portfolio selection model is proposed. The proposed model covered minimum transaction lots, cardinality constraints and also regards market (sector) capitalization. A Genetic Algorithm which is coded by MATLAB 7 software is produced to solve the proposed model which belongs to mixed-integer nonlinear programming (NP-Hard). The empirical studies first compare the proposed GA and the model results in LINGO; in addition the proposed model is verified utilizing a market with 500 and 2000. The first experiment indicates the difference between LINGO's global optimum and GA's best objective is just 2.9%. Such a small difference which is below than 3% can be acceptable and reliable. The second experiment shows the appreciable performance of proposed GA in achieving to an acceptable objective in a less than 7 min with just 3.5% of risk (it assures 97.5% of confidence limits to the investor). Our experiments can show that this model and the proposed GA are applicable and reliable in real markets with large number of stocks (see Table 6).

Finally, for future researches four areas are proposed; first adding other constraints of real market like Transaction costs, second changing the proposed model to a multi period one, third using other heuristic algorithms to solve the proposed model and comparing their solutions with GA and at last, verifying the performance of proposed model in a market.

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